NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 12

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

In an ICMP Echo sweep, if the scanner receives an ICMP Type 0 packet in response to an ICMP Type 8 request, what does this indicate?

a. The target host is alive/up
b. The target host is down
c. The target port is filtered
d. The target host does not support ICMP

Correct Answer: a

Detailed Solution: ICMP Type 0 is an Echo Reply. Receiving this in response to an Echo Request (Type 8) confirms that the target host is alive and responding.
Thus, the correct option is (a).

QUESTION 2:

Which of the following NMAP option(s) can be used to perform a TCP-based sweep for host discovery?

a. -PE
b. -PS
c. -PA
d. -PU
e. -PM

Correct Answer: b, c

Detailed Solution: -PS (TCP SYN ping) and -PA (TCP ACK ping) are used for TCP-based host discovery sweeps. -PE is ICMP Echo sweep, -PU is UDP sweep, and -PM is ICMP Address Mask (non-echo) sweep.
Thus, the correct options are (b) and (c).

QUESTION 3:

Which of the following are automatically performed when using the NMAP -sn option?

a. ICMP Echo sweep
b. TCP sweep
c. UDP sweep
d. All of these

Correct Answer: a, b

Detailed Solution: The -sn option performs host discovery using ICMP Echo and TCP sweep techniques together, but does not include UDP sweep as part of its default behavior.
Thus, the correct options are (a) and (b).

QUESTION 4:

Which NMAP option treats all hosts as online, effectively skipping the host discovery phase?

a. -sL
b. -sP
c. -PO
d. -sU
e. -Pn

Correct Answer: e

Detailed Solution: The -Pn option instructs NMAP to skip host discovery entirely and treat all specified hosts as online, proceeding directly to the scanning phase. -sL only lists targets, -sP performs a ping-only scan, -PO is IP protocol ping, and -sU performs a UDP scan.
Thus, the correct option is (e).

QUESTION 5:

Consider the following statements:

(i) -sn performs host discovery only, without any port scanning.

(ii) -Pn skips host discovery entirely and proceeds directly to port scanning.

a. Only (i) is true
b. Only (ii) is true
c. Both (i) and (ii) are true
d. Both (i) and (ii) are false

Correct Answer: c

Detailed Solution: -sn performs host discovery only, skipping port scanning entirely, while -Pn does the opposite — skipping host discovery and assuming all hosts are online, which is useful when a target blocks ICMP-based discovery but is still reachable for port scanning. These two options serve distinct, near-opposite purposes, not identical ones, and -sn does not perform a port scan afterwards.
Thus, the correct option is (c).

QUESTION 6:

Which of the following statement(s) is/are **true** regarding NMAP's -F (fast scan) option?

a. It scans the top 100 most common ports.
b. It scans all 65535 ports.
c. It is faster than a default scan because it checks fewer ports.
d. It is equivalent to using --top-ports 1000

Correct Answer: a, c

Detailed Solution: The -F (fast) option scans only the top 100 most common ports, making it quicker than the default scan (which covers the top 1000 ports). It does not scan all 65535 ports, and it is not equivalent to --top-ports 1000, since it covers only 100 ports, not 1000.

Thus, the correct options are (a) and (c).

QUESTION 7:

Which NMAP option is specifically used to conduct a UDP port scan?

a. -sT
b. -sS
c. -sU
d. -sA
e. -sN

Correct Answer: c

Detailed Solution: The -sU option performs a UDP scan. -sT is TCP Connect scan, -sS is TCP SYN scan, -sA is TCP ACK scan, and -sN is NULL scan (all TCP-based, not UDP).

Thus, the correct option is (c).

QUESTION 8:

Which scanning technique completes the full TCP three-way handshake with the target port to determine its state?

a. TCP SYN scan
b. TCP Connect scan
c. FIN scan
d. NULL scan
e. Xmas scan

Correct Answer: b

Detailed Solution: TCP Connect scan (-sT) completes the entire three-way handshake (SYN, SYN-ACK, ACK) with the target port. TCP SYN scan is a "half-open" scan that doesn't complete the handshake, while FIN, NULL, and Xmas scans use unusual flag combinations and don't involve a standard handshake at all.

Thus, the correct option is (b).

QUESTION 9:

How many ports will be scanned using the NMAP command "nmap --top-ports 20 target_ip"?

Correct Answer: 20

Detailed Solution: The --top-ports option scans the N most commonly used ports specified. Here, N = 20, so exactly 20 of the most common ports will be scanned.

QUESTION 10:

How many total IP addresses will be scanned when executing the command nmap -sL 10.10.5.100-115?

Correct Answer: 16

Detailed Solution: The command scans all IP addresses from 10.10.5.100 to 10.10.5.115, inclusive.
Thus, the correct answer is 16.

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